

NATHANIEL MCINTIRE

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EDUCATION

NORTH CAROLINA STATE UNIVERSITY

Expected Graduation: May 2027

Bachelor of Aerospace Engineering | Minor Astrophysics | Minor Mathematics

- Courses: Aerodynamics I & II, Aerospace Structures I & II, Intro. to Space Flight, Modern Physics, Astrophysics, Automatic Controls, Thermodynamics, Applied Machine Learning, Aerospace Vehicle Performance, MATLAB, SolidWorks

BRUNSWICK COMMUNITY COLLEGE

May 2024

Associates of Engineering | Cum Laude Honors

BRUNSWICK COMMUNITY COLLEGE

May 2022

Associates of Science | Cum Laude Honors

PROFESSIONAL EXPERIENCE

BONGIORNO & SON ITALIAN SPECIALTIES

March 2025 - Present

Supervisor

Raleigh, NC

- Supervised daily front and back-of-house staff for a team of 10 employees, ensuring efficient workflow, food quality, and guest satisfaction.
- Manage opening and closing procedures, cash handling, inventory control, and scheduling in a fast-paced airport restaurant environment.
- Resolve guest concerns professionally while maintaining high service standards and maintaining our 5/5-star rating on Google.
- Reduced labor costs by 10%, while increasing average sales by 25% over 3 months.
- Oversee daily sales operations totaling \$3k-\$6.5k per day.

ACE HARDWARE

August 2024 - March 2025

Supervisor

Raleigh, NC

- Managed daily store operations for a team of 10 employees, including inventory management, cash handling, guest satisfaction, and appliance assembly.
- Processed and organized approximately 10-20 customer deliveries weekly, as well as ordering store shipments twice a week and maintaining accurate inventory and store readiness.
- Operated forklift equipment to unload trucks and move pallets weighing upwards of 1200 lbs.
- Maintained a positive shopping experience for guests by assisting with specific requests and helping them with their exact needs.
- Assembled approximately 20 appliances a week, including grills, lawn mowers, and chainsaws, for guest pickup, deliveries, or displays.

LOWES FOODS

October 2020 - August 2024

Guest Services Manager

Wilmington, NC

- Managed front-end operations for a team of 40 employees, including cashiers, guest services, cash office, Lowes Foods To-Go, and facilities maintenance, which included scheduling, cash handling, supply ordering, and guest services.
- Oversaw all lottery, Western Union, and guest service transactions, which could total upwards of 20k per day.
- Conducted interviews, onboarding, drug testing, and orientation for 3 new hires per week.
- Led the completion of training modules for all new hires in compliance with company policy.
- Improved promotional sales by 10% while reducing transactional errors and maintaining operational efficiency.
- Earned a 95% audit score and consistently earned 100% secret shopper scores.
- Responsible for the secure collection of all transactions, totaling about 60-80k per day.

ENGINEERING PROJECTS

HIGH-POWERED ROCKETRY CLUB

- Experience with KiCAD and PCB design, as well as hands-on experience with carbon-fiber layups and payload and ejection tests.

AERODYNAMICS II

- Air Flow Analysis through a CD Nozzle - Designed and manufactured a CD nozzle using SolidWorks and ANSYS for the CFD model, then used NCSU's Wind Tunnel to gather data points, and then used MATLAB to process and plot the data.
- Analyzed an Impinging Jet Flow using PSP - Used a binary PSP to gather the pressure distribution over a flat surface, then used image processing in MATLAB to analyze the data.

AEROSPACE STRUCTURES II

- Composite Laminate Stress Analysis - Performed stress evaluations for composite laminate plies across various loads, facilitating the transformation of stress values between material and global coordinate systems.

AUTOMATIC CONTROLS

- Lead Compensator Design for a Pendulum System - Developed a lead compensator for a linearized pendulum model, utilizing root locus techniques to achieve target performance metrics for overshoot and settling time.
- Used ladder logic to control a PLC to pick up an object and transfer it to another location.

APPLIED MACHINE LEARNING

- Comparison of Different Machine Learning Techniques to Predict and Model Flight Delays - Engineered Gini-Based CART, Entropy-Based CART, and Random Forest models; utilized SMOTE and Tomek Link techniques for dataset optimization and conducted a comparative analysis of algorithmic efficacy.

TECHNICAL SKILLS

MATLAB | Python | C++ | HTML | FEA | ANSYS | KiCAD | AutoCAD | SolidWorks | Machine Learning | Numerical Analysis | Orbital Mechanics | CFD | Simulink | PID Design | Root Locus Design | Propulsion Systems | Structural Analysis | Neural Networks | Lead Compensation | Ladder Logic | Project Management | XFOIL | Materials | Quantum Mechanics | Semiconductor Design | Astrophysics | PLC Control